

Retail Business Performance & Profitability Analysis

Prepared by: Aishwarya — **Tools:** SQL · Python · Looker Studio — **Dataset:**
Superstore Sales Data (9,994 Records)

Introduction

Retail businesses often struggle to identify which product categories, regions, and seasons are driving or draining profitability. Without clear data-driven insights, businesses risk overstocking unprofitable items, ignoring underperforming regions, and offering discounts that hurt margins more than they help sales.

This project analyzes transactional retail data from the globally recognized Superstore dataset to uncover profit-draining categories, examine regional performance gaps, identify seasonal sales patterns, and derive actionable strategic recommendations. The analysis follows an end-to-end data analytics workflow — from raw data cleaning in SQL, through Python-based statistical analysis and visualization, to an interactive Looker Studio dashboard.

Tools Used

- **SQL (CSVFiddle):** Data import, null value checks, profit margin calculations by Category, Sub-Category, Region, and monthly seasonal aggregations using STRFTIME and GROUP BY
- **Python (Google Colab):** Pandas for data loading and manipulation, Seaborn and Matplotlib for bar charts, horizontal bar charts (red for losses), multi-year line charts, and discount vs profit scatter plot correlation analysis
- **Looker Studio (Google Data Studio):** Interactive dashboard featuring profit by category bar chart, profit by region bar chart, sales by customer segment pie chart, sub-category profit ranking table, and two dropdown filters for Region and Category

Steps Involved in Building the Project

1. **Data Collection:** Downloaded the Superstore Sales dataset from Kaggle — 9,994 rows and 21 columns covering orders from 2014 to 2017 across the United States
2. **Data Import & Cleaning (SQL):** Uploaded CSV to CSVFiddle and ran COUNT checks on Sales, Profit, and Category columns — confirmed zero null values and 100% data completeness
3. **Profit Margin Analysis (SQL):** Used SUM and GROUP BY to calculate total sales, total profit, and profit margin percentage at both Category and Sub-Category levels — identified Tables, Bookcases, and Supplies as loss-making
4. **Regional Analysis (SQL):** Aggregated profit margins by Region — identified Central as the weakest region at 7.92% and West as the strongest at 14.94%

5. **Seasonal Analysis (SQL):** Used STRFTIME to extract month from Order Date and grouped sales and profit by month — confirmed Q4 as peak season and February as weakest month
6. **Python Visualizations (Google Colab):** Loaded dataset using Pandas, built category profit margin bar chart, sub-category horizontal bar chart with red bars for losses, multi-year seasonal line chart, and discount vs profit scatter plot with correlation coefficient
7. **Dashboard (Looker Studio):** Connected Superstore CSV as data source, built 4 interactive charts, added Region and Category dropdown filters for dynamic filtering across all visuals
8. **Strategic Recommendations:** Derived business recommendations for loss-making items, regional underperformance, discount capping, and Q4 inventory planning

Conclusion

This project successfully demonstrates an end-to-end retail business performance analysis using industry-standard data analytics tools. The analysis reveals that the Superstore business is significantly held back by loss-making Furniture sub-categories, underperforming Central region operations, and counterproductive discounting practices.

Key recommendations include: discontinuing or repricing Tables and Bookcases, capping all discounts at a maximum of 20%, building Q4 inventory by September to capitalize on peak demand, investigating and replicating West region strategies across Central and South, and prioritizing Technology category expansion given its superior 17.4% profit margin. Implementing these recommendations could meaningfully improve overall profitability without requiring an increase in total sales volume — demonstrating the power of data-driven decision making in retail operations.